Item [Options]	**Quantity**	**Cost per pack**	**Link**	**How many do we need for testing?**	**PROs**	**CONs**
Microcontrollers						
-Nano					simplest, Aruino	needs transceiver
ELEGOO Nano Board CH340/ATmega328P Without USB Cable, Compatible with Arduino Nano V3.0 (Nano x 3 Without Cable)	3/pack	$19.99	https://www.ama	2 packages would allow for testing of 2 boards w/sensors and 1 board w/transceiver per hardwa	no wires, header pins not soldered	
Nano V3.0, Nano Board ATmega328P 5V 16M Micro-Controller Board Compatible with Arduino IDE (Nano x 3 with USB Cable)	3/pack	$13.99	Amazon.com: Na	2 packages would allow for testing of 2 boards w/ser	-has wires, header pins attached	
-NodeMCU					built-in WiFi, popular for IoT projects, can u	maybe harder to use, not Arduino language
HiLetgo 3pcs ESP8266 NodeMCU CP2102 ESP-12E Development Board Open Source Serial Module Works Great for Arduino ID	3/pack	$13.99	Amazon.com: Hil	2 packages would allow for testing of 2 boards w/sensors and 1 board w/transceiver per hardware developer		
MakerFocus 3pcs ESP8266 NodeMCU LUA CP2102 ESP-12E Internet WiFi Development Board Serial Wireless Module Internet	3/pack	$13.99	Amazon.com: MakerFocus 2pcs ESP8266 NodeMCU LUA CP2102 ESP-12E Internet WiFi Development Board Serial Wireless Module Internet for Ar duino IDE/Micropython with Free Adapter Board for ESP8266 ESP-01 and nRF24L01+: Computers & Accessories			
KeeYees Development Board WiFi WLAN Wireless Module for ESP8266 for NodeMCU for ESP-12E for Arduino (3pc)	3/pack	$11.99	Amazon.com: KeeYees Development Board WiFi WLAN Wireless Module for ESP8266 for NodeMCU for ESP-12E for Arduino (3pc): Computers & Accessories			
Sensors						
Geekstory HC-SR501 Pir Pyroelectric Infrared PIR Sensor Body Motion Sensor Modules for Arduino (Pack of 5pcs)	5/pack	$8.49	https://www.ama	1 package would allow for testing of 2 sensors per hardware developer (with 1 extra for future use)		
Transceivers						
HiLetgo 2pcs NRF24L01+PA+LNA Wireless Transceiver RF Transceiver Module with SMA Antenna 2.4G 1100m	2/pack	$8.99	https://www.ama	1 package would allow for testing of all sensors per hardware developer		
Makerfire 10pcs Arduino NRF24L01+ 2.4GHz Wireless RF Transceiver Module New	10/pack	$12.88	Amazon.com: Makerfire 10pcs Arduino NRF24L01+ 2.4GHz Wireless RF Transceiver Module New: Electronics			
-Adapter						
Wires						
Female-to-Female Jumper Wire	40/pack	$4.59	Amazon.com: An	1 package		
Female-to-Female Jumper Wire	40/pack	$4.99	Amazon.com: Ge	1 package		
Cable						
1000ft spool of 22 AWG 3/C Cable	1000ft	$126.80	https://www.ama	1 package		
1000ft spool of 22 AWG 3/C Cable	1000ft	$126.80	https://www.ama	1 package		
1000ft spool of 22 AWG 3/C Shielded Cable	1000ft	$166.80	https://www.ama	1 package		
22 AWG 3/C Shielded Cable	by the foot	$0.18	https://skycraftsu	~1000		
1000ft spool of 22 AWG 3/C Shielded Cable	1000ft	$110.00	https://www.wire	1 package		
24 AWG / 3 Pair Unshielded	by the foot	$0.14	https://www.platt	~1000		
22 AWG 4/C Cable	200ft	$27.95	https://www.ebay	~5		
24 AWG 3 conductor wire Power Cable	33ft	$13.99	https://www.ama	~30		
Breadboards/Protoboards/PCBs						
HiLetgo 20pcs 5x7cm Bakelite DIY Prototype Board PCB	20/pack	$6.49	https://www.ama	4 packages		
KeeYees 2pcs Expansion Board Kits GPIO 1 into 2 for ESP8266 ESP-12E NodeMCU WiFi Development Board CP2102	2/pack	$8.99	https://www.ama	~32 packages		
CircuitSetup Solderable Breadboard PCB for Electronics, RaspberryPi, ESP8266, ESP32 Prototyping - 3.78"x2.2" (96x56mm) (5	5/pack	$11.99	https://www.ama	13 packages		
Connectors/Headers/Terminals						
Housing						
Power Supply						
Soldering Equipment						
Questions: Do we need wires to connect these all together? Are there other hardware pieces required to make things connect correctly? How are we going to power everything?						
(Mark S.) addressing the above questions:						
-it seems like the header pins are included with the ELEGOO boards in the link above, but it's sort of unclear. Either way, the pins are unsoldered, so we need to do that ourselves. Personally I have what I need to do that, but solder should probably be added to the list						
-we definitely need wires. For development, female-female and/or male-female jumper wires are probably easiest, because it seems like all the products on our list have jumper pins. Breadbords will be helpful for development too. PCB protoboards and regular single conductor wire would probably be good too but that involves soldering						
-we should look into 5v power supplies that should be enough to power multiple of these devices. The RF transceiver has a breakout adapter and 3.3v regulator that we can buy, and that might be a good idea. I don't know if the 3.3v pin on the Nano can provide enough power						
-the easiest thing to do would probably be to get a power supply for each room that can plug into a regular outlet, but we can probably save resources by sharing between rooms. I was thinking we might not even need a Nano in every room. This all depends on how we can route wires between rooms though						
-I will put together links and stuff for a lot of this info and add to this doc. I'm just dumping my thoughts here for now						
-another thing to note is Amazon is good and has what we need and is priced well, but Mouser.com and Digikey.com are the most popular resources for electronics so don't forget to look there too. At least to check prices (especially for buying in bulk)						